

IA na Saúde Mental: Predição de Burnout

Modelo Preditivo para identificar colaboradores em risco de esgotamento



Núcleo de Capacitação em Inteligência Artificial

Discentes:

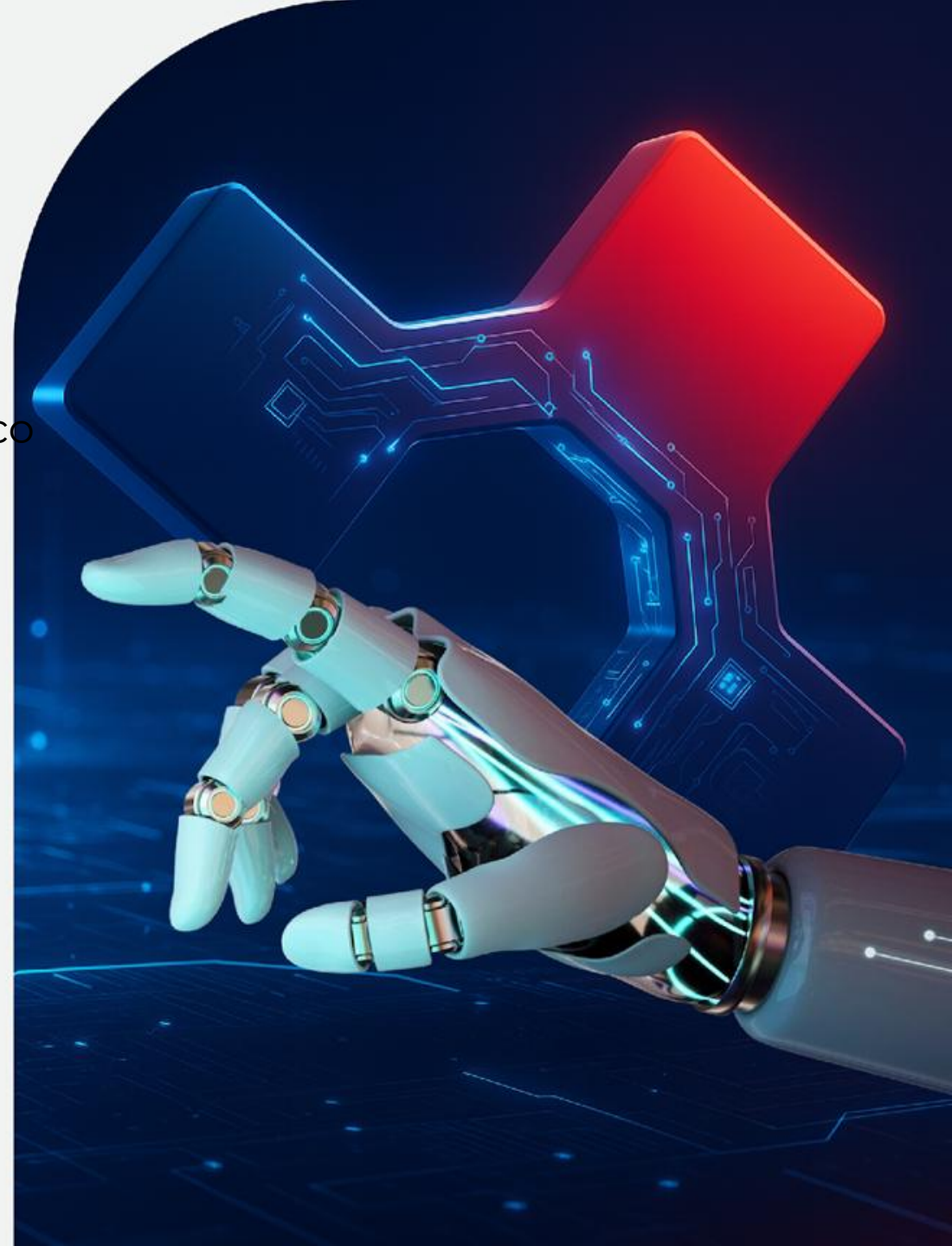
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Viviane Dos Santos Amorim



Kaggle Dataset

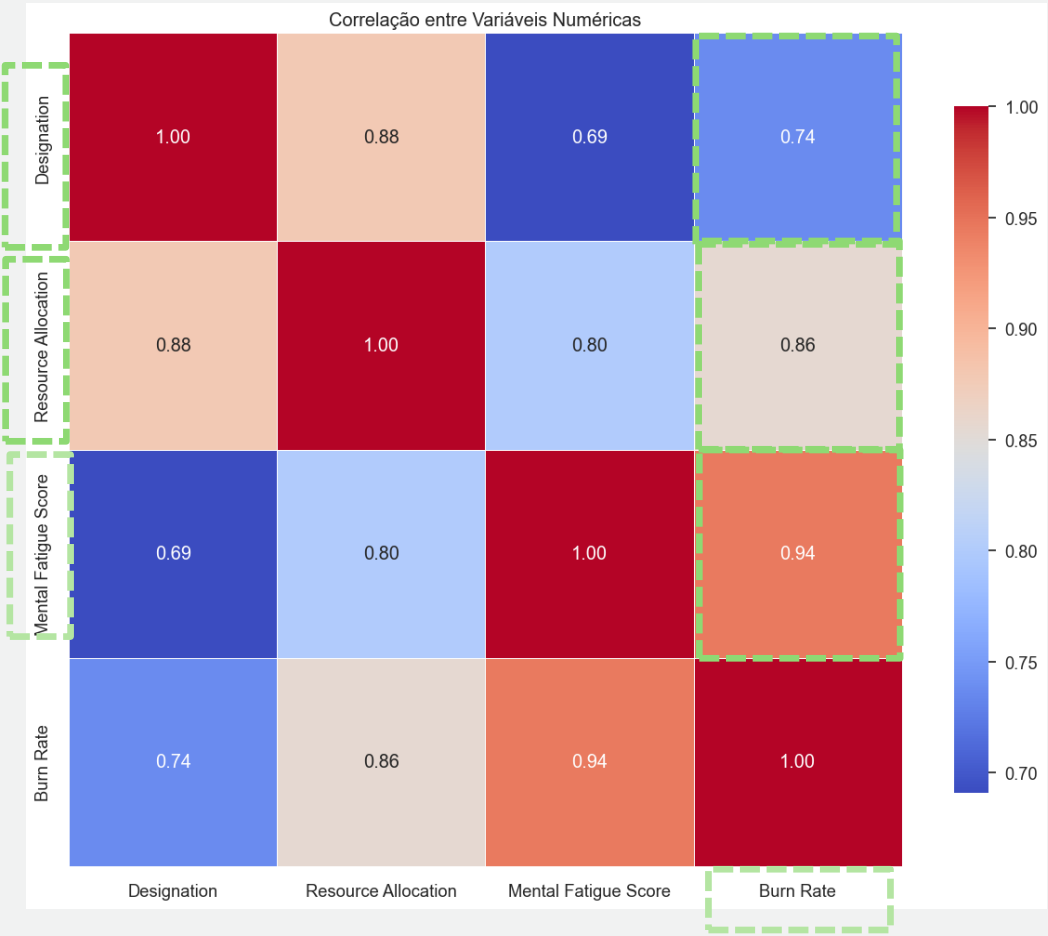
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 22750 entries, 0 to 22749
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Employee ID           22750 non-null  object
1   Date of Joining       22750 non-null  object
2   Gender                22750 non-null  object
3   Company Type          22750 non-null  object
4   WFH Setup Available   22750 non-null  object
5   Designation           22750 non-null  float64
6   Resource Allocation    21369 non-null  float64
7   Mental Fatigue Score  20633 non-null  float64
8   Burn Rate             21626 non-null  float64
dtypes: float64(4), object(5)
memory usage: 1.6+ MB
```

Dataset		
Atributos	Kaggle	Havard
Categoricos	5	5
Numéricos	4	12
Total	9	17

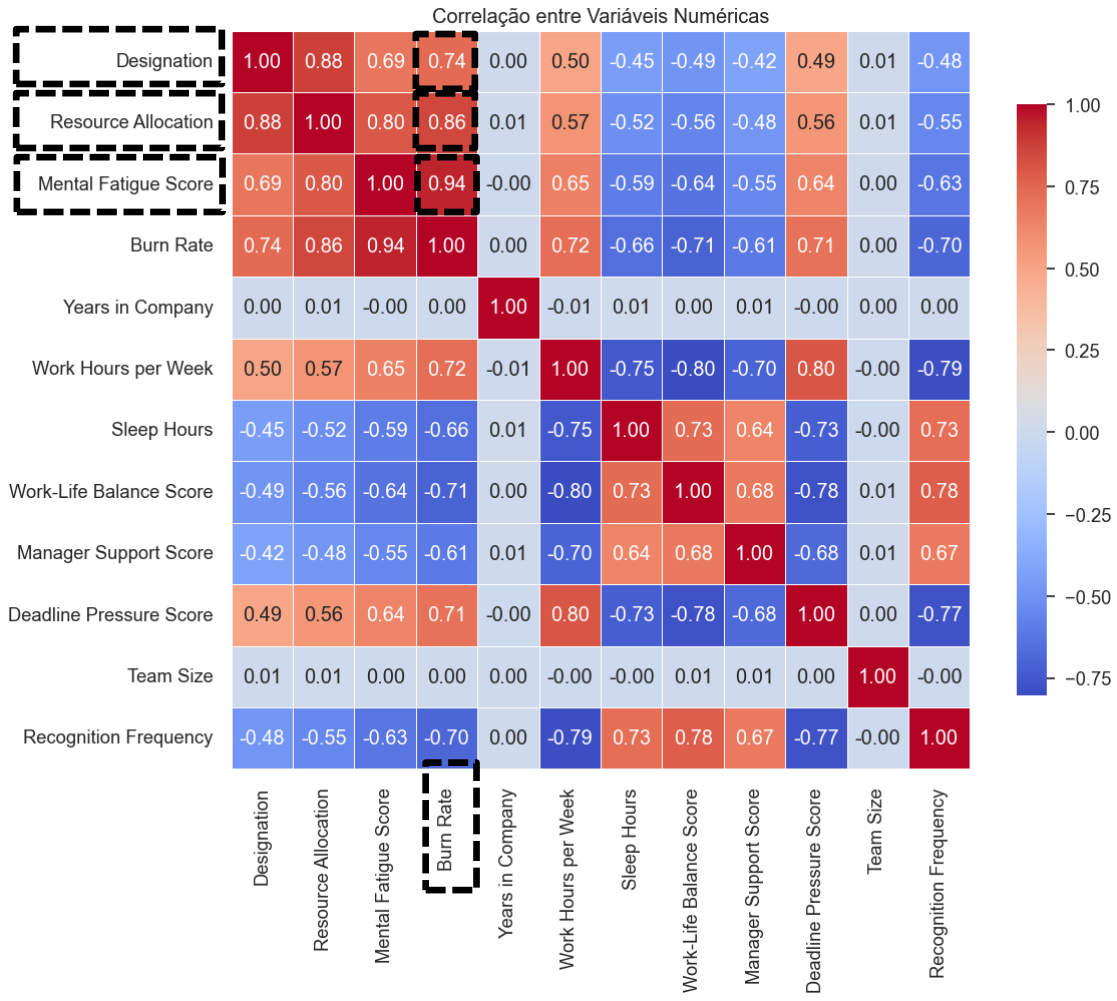
Havard Dataset

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 22750 entries, 0 to 22749
Data columns (total 17 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Employee ID           22750 non-null  object
1   Date of Joining       22750 non-null  object
2   Gender                22750 non-null  object
3   Company Type          22750 non-null  object
4   WFH Setup Available   22750 non-null  object
5   Designation           22750 non-null  int64
6   Resource Allocation    21369 non-null  float64
7   Mental Fatigue Score  20633 non-null  float64
8   Burn Rate             21626 non-null  float64
9   Years in Company      22750 non-null  int64
10  Work Hours per Week   22750 non-null  int64
11  Sleep Hours           22750 non-null  float64
12  Work-Life Balance Score 22750 non-null  int64
13  Manager Support Score 22750 non-null  int64
14  Deadline Pressure Score 22750 non-null  int64
15  Team Size             22750 non-null  int64
16  Recognition Frequency 22750 non-null  int64
dtypes: float64(4), int64(8), object(5)
memory usage: 3.0+ MB
```

Kaggle Dataset



Havard Dataset



 Melhoria por Modelo (Harvard Dataset em relação ao Kaggle)

Modelo	% Melhoria RMSE	% Melhoria MAE	% Melhoria R²
GradientBoostingRegressor	13,40%	13,96%	2,59%
StackingRegressor	13,18%	13,33%	2,56%
RandomForest	13,53%	14,01%	2,67%
DT	6,70%	8,86%	1,46%
Linear	8,83%	8,61%	2,57%
Lasso	8,83%	8,62%	2,57%
Média	10,74%	11,23%	2,40%

Kaggle Dataset

	Modelo	CV_RMSE	Test_RMSE	Test_MAE	Test_R2
0	GB	0.060469	0.059746	0.046869	0.906280
1	Stacking	0.060543	0.059861	0.046929	0.905922
2	RF	0.061191	0.060376	0.047148	0.904295
3	DT	0.063328	0.062088	0.048443	0.898790
4	Linear	NaN	0.070927	0.053461	0.867923
5	Lasso	0.071452	0.070932	0.053459	0.867905

Havard Dataset

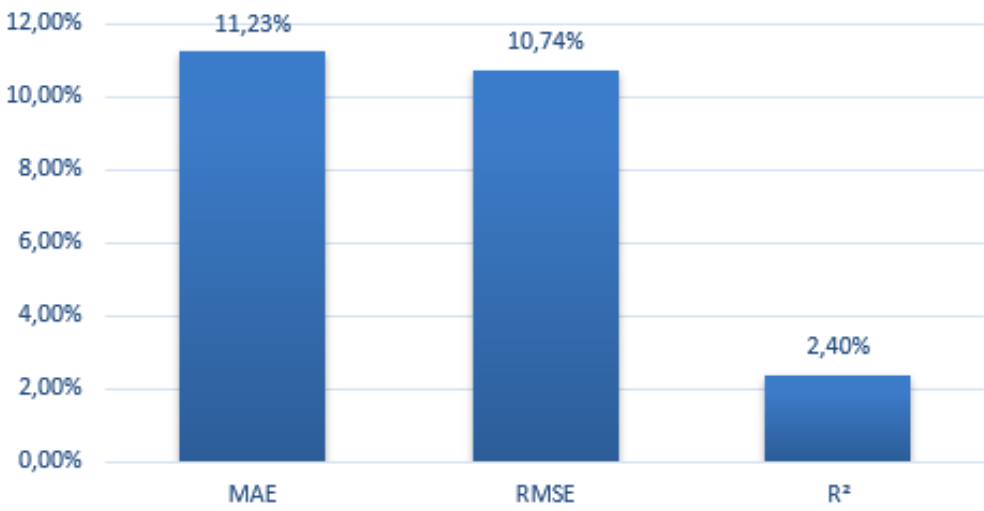
	Modelo	CV_RMSE	Test_RMSE	Test_MAE	Test_R2
0	GB	0.052656	0.051740	0.040326	0.929716
1	Stacking	0.052896	0.051973	0.040675	0.929081
2	RF	0.053188	0.052206	0.040543	0.928445
3	DT	0.058783	0.057925	0.044150	0.911907
4	Linear	NaN	0.064667	0.048856	0.890206
5	Lasso	0.065274	0.064669	0.048851	0.890202

 Resumo Geral

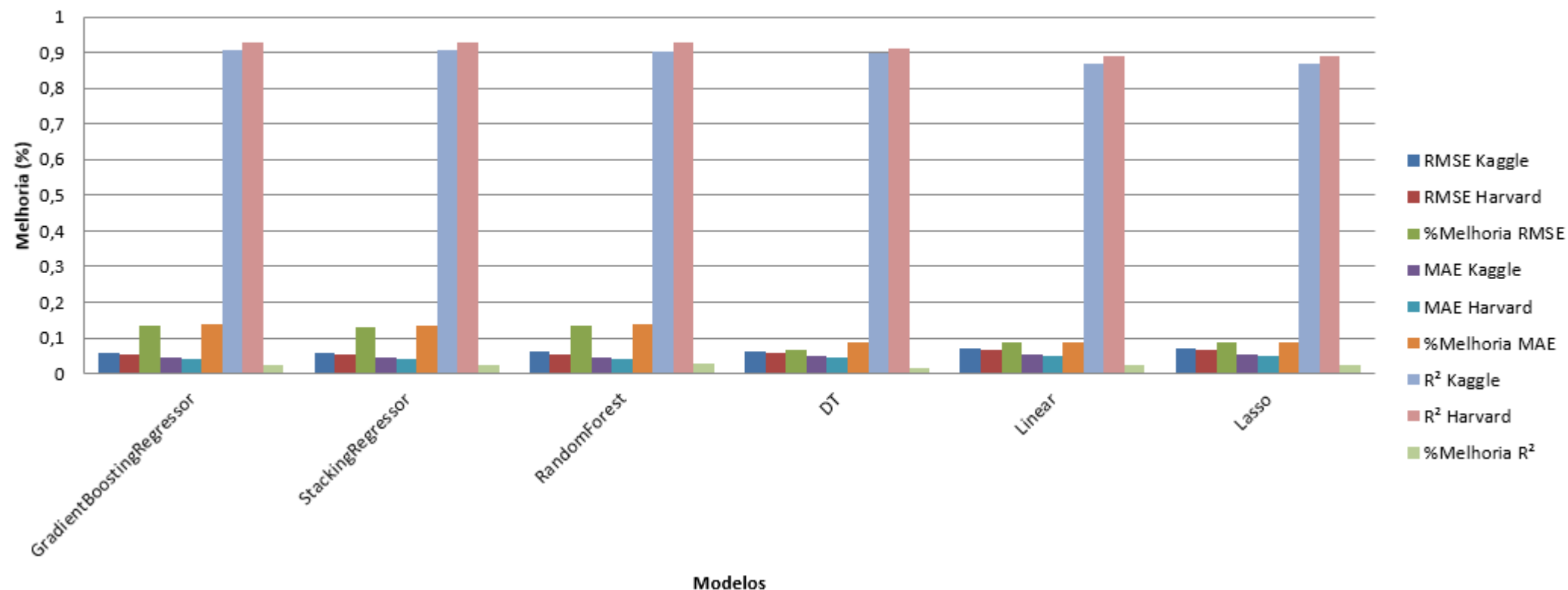
O modelo treinado com o **Harvard Dataset** apresentou melhorias médias de:

Métrica	Melhoria Média (%)	Interpretação
MAE	11,23%	O erro absoluto médio foi 11,23% menor (menor diferença entre predições e valores reais).
RMSE	10,74%	O erro quadrático médio foi, em média, 10,74% menor.
R²	2,40%	O poder explicativo aumentou cerca de 2,40%.

Melhoria Média (%) - Harvard vs Kaggle



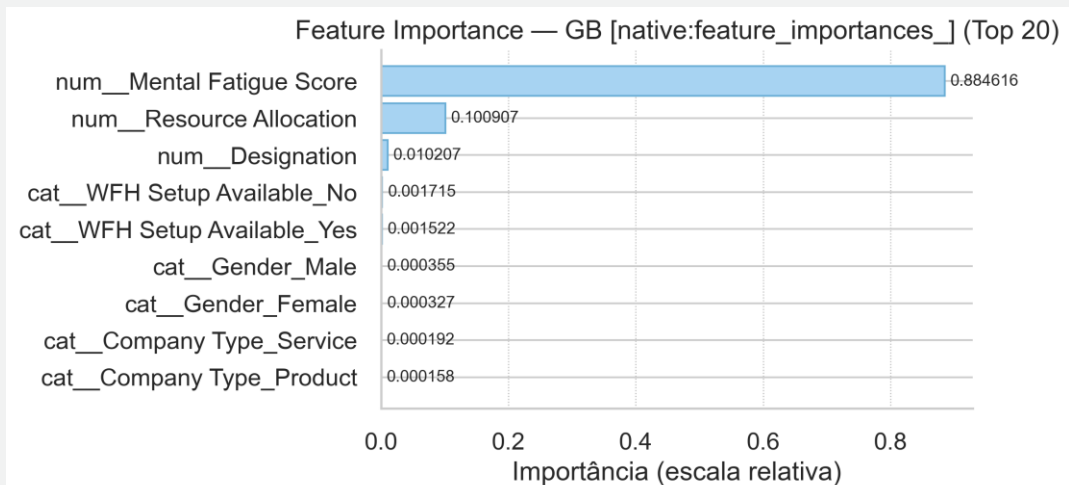
Melhoria (%) Harvard vs Kaggle



O gráfico comparativo de desempenho está mostrando o quanto o **Harvard Dataset** supera o **Kaggle Dataset** em cada métrica:

- **Barras verdes** → melhoria em **RMSE** (quanto menor, melhor).
- **Barras laranja** → melhoria em **MAE** (quanto menor, melhor).
- **Barras verdes claros** → melhoria em **R²** (quanto maior, melhor).

Kaggle Dataset



Havard Dataset

